

Extensions of the CQC Counterexample

Higher dimensions, the two-qubit case, and replacement upper bounds

Working research note – July 2026

Status and provenance. This note develops consequences of the explicit qutrit-pair counterexample to the complementary-quantum correlation (CQC) conjecture. The original qutrit calculation has been independently reproduced outside the model-assisted workflow. The higher-dimensional computational/Fourier family is algebraic and has been checked directly in the full d^2 -dimensional space for $2 \leq d \leq 10$. The arbitrary-MUB-pair extension and the conclusions about unrestricted two-qubit states should receive independent scrutiny before being treated as settled theorems.

The analysis and initial verification artifacts were generated by OpenAI GPT-5.6 Pro in a prompted research exchange; Ethan O'Connor is the repository steward. Source, exact qutrit certificate, and verification code: github.com/EthanOConnor/cqc-qutrit-counterexample.

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1 Summary of the main conclusions

The qutrit construction is not isolated. A rank-two family exists in every equal local dimension $d \geq 2$ for the computational and Fourier mutually unbiased bases (MUBs), with exact CQC gap

$$\Delta_d(\epsilon) = h_2(\epsilon) - 2h_2(\epsilon/d). \quad (1)$$

At $\epsilon = 1/10$, this family violates CQC for every $d \geq 3$, and the violation increases with d . The same family has an exact no-go at $d = 2$.

No unrestricted two-qubit counterexample has been found in the searches reported here. The original large random search, the known proved cases, and supplementary exploratory searches all point toward the possibility that the qubit case survives as a separate theorem, but no general proof is supplied here [1].

Even after CQC fails, several universal replacement bounds remain. For two local MUB pairs,

$$I_Z + I_X \leq \min\{\log_2 d - H(A | B), \log_2 d - H(B | A)\}. \quad (2)$$

For m pairwise MUB measurement pairs,

$$\sum_{i=1}^m I(M_i^A : M_i^B) \leq \frac{m}{2} \min\{\log_2 d - H(A | B), \log_2 d - H(B | A)\}. \quad (3)$$

These bounds also yield useful entropy- and purity-dependent corollaries.

2 An all-dimension rank-two family

2.1 Logical qubit inside $\mathbb{C}^d$

For $d \geq 2$, define

$$|0_L\rangle = |0\rangle, \quad |1_L\rangle = \frac{1}{\sqrt{d-1}} \sum_{j=1}^{d-1} |j\rangle, \quad (4)$$

and set

$$c_d = \frac{1}{\sqrt{d}}, \quad s_d = \sqrt{\frac{d-1}{d}}. \quad (5)$$

Define two orthogonal, maximally entangled states on this logical two-dimensional support:

$$|\phi_d\rangle = \frac{|0_L 0_L\rangle + |1_L 1_L\rangle}{\sqrt{2}}, \quad (6)$$

$$|\chi_d\rangle = \frac{1}{\sqrt{2}} [s_d(|0_L 0_L\rangle - |1_L 1_L\rangle) - c_d(|0_L 1_L\rangle + |1_L 0_L\rangle)]. \quad (7)$$

Mix them as

$$\rho_{d,\epsilon} = (1 - \epsilon)|\phi_d\rangle\langle\phi_d| + \epsilon|\chi_d\rangle\langle\chi_d|, \quad 0 \leq \epsilon \leq 1. \quad (8)$$

For $d = 3$, this is the qutrit counterexample already established.

Both components have maximally mixed logical marginals, so

$$\rho_A = \rho_B = \frac{1}{2}\Pi_L, \quad (9)$$

where Π_L projects onto $\text{span}\{|0_L\rangle, |1_L\rangle\}$. The nonzero eigenvalues of $\rho_{d,\epsilon}$ are $1 - \epsilon$ and ϵ , hence

$$I(A : B) = 2 - h_2(\epsilon). \quad (10)$$

2.2 Computational-basis mutual information

After coarse-graining physical outcomes $1, \dots, d-1$ into one logical outcome, the computational measurement has binary joint distribution

$$P_{\text{logical}} = \begin{pmatrix} \frac{1}{2} - \frac{\epsilon}{2d} & \frac{\epsilon}{2d} \\ \frac{\epsilon}{2d} & \frac{1}{2} - \frac{\epsilon}{2d} \end{pmatrix}. \quad (11)$$

To see explicitly why the full d -outcome measurement has the same mutual information, write

$$a = \frac{1}{2} - \frac{\epsilon}{2d}, \quad b = \frac{\epsilon}{2d}. \quad (12)$$

Its fine-grained probabilities are

$$P_{00} = a, \quad (13)$$

$$P_{0k} = P_{k0} = \frac{b}{d-1}, \quad k > 0, \quad (14)$$

$$P_{jk} = \frac{a}{(d-1)^2}, \quad j, k > 0. \quad (15)$$

Thus, conditioned on each coarse logical outcome, the physical labels are generated by independent local uniform refinements. Such refinements preserve mutual information. Therefore

$$I_Z = 1 - h_2(\epsilon/d). \quad (16)$$

2.3 Fourier-basis mutual information

Let

$$|x_k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{jk} |j\rangle, \quad \omega = e^{2\pi i/d}, \quad (17)$$

be the Fourier basis. Its projectors compressed to the logical support are

$$\Pi_L |x_0\rangle\langle x_0| \Pi_L = |u\rangle\langle u|, \quad (18)$$

$$\Pi_L |x_k\rangle\langle x_k| \Pi_L = \frac{1}{d-1} |v\rangle\langle v|, \quad k > 0, \quad (19)$$

where

$$|u\rangle = c_d |0_L\rangle + s_d |1_L\rangle, \quad |v\rangle = s_d |0_L\rangle - c_d |1_L\rangle. \quad (20)$$

Indeed, $\sum_{j=1}^{d-1} \omega^{jk} = d-1$ for $k=0$ and -1 for $k>0$. Hence the Fourier measurement is the logical $\{|u\rangle, |v\rangle\}$ measurement, with the v outcome split uniformly among $d-1$ physical labels.

Let

$$O_d = \begin{pmatrix} c_d & s_d \\ s_d & -c_d \end{pmatrix}. \quad (21)$$

A direct calculation gives

$$(O_d \otimes O_d) |\phi_d\rangle = |\phi_d\rangle, \quad (O_d \otimes O_d) |\chi_d\rangle = -|\chi_d\rangle. \quad (22)$$

The density operator is invariant under the logical basis change, so the logical joint table is the same as in the computational basis. The fine physical labels are again independent uniform refinements, and therefore

$$I_X = I_Z = 1 - h_2(\epsilon/d). \quad (23)$$

2.4 Exact CQC gap and dimensional threshold

Combining the preceding formulas gives

$$\Delta_d(\epsilon) := I_Z + I_X - I(A : B) = h_2(\epsilon) - 2h_2(\epsilon/d). \quad (24)$$

At $\epsilon = 1/10$:

d	$\Delta_d(1/10)$ (bits)
2	-0.103798320642631
3	+0.047310992952217
4	+0.131673730595941
5	+0.186114508505640
6	+0.224412399450532
10	+0.307409321797459

For $d = 3$ this is the qutrit gap. Since $h_2(1/(10d))$ decreases with $d \geq 3$, the violation grows for higher dimensions. An exact integer sign certificate is

$$10d \Delta_d(1/10) = \log_2 \left[\frac{(10d-1)^{20d-2}}{10^{10d} d^{20d} 3^{18d}} \right]. \quad (25)$$

Thus

$$\Delta_d(1/10) > 0 \iff (10d-1)^{20d-2} > 10^{10d} d^{20d} 3^{18d}. \quad (26)$$

At $d = 3$ this reduces to $29^{58} > 3^{114} 10^{30}$.

Because the gap is strictly positive, sufficiently small white-noise perturbations

$$(1-\eta)\rho_{d,1/10} + \eta \frac{I_{d^2}}{d^2} \quad (27)$$

are full-rank counterexamples for every $d \geq 3$ by continuity.

2.5 Dependence on the chosen MUB pair

The computational/Fourier derivation extends to any prescribed pair of rank-one MUBs. The transition matrix between two MUBs is a complex Hadamard matrix. Rephase rows and columns so its first row and first column are all $1/\sqrt{d}$. Orthogonality then implies that, in every nonfirst column, the sum of the remaining entries is $-1/\sqrt{d}$.

Choose $|0_L\rangle$ as the first vector of the first basis and $|1_L\rangle$ as the normalized uniform sum of the remaining vectors. The second basis then has exactly the same compressed projectors as above:

$$|u\rangle\langle u|, \quad \frac{1}{d-1}|v\rangle\langle v|, \dots, \frac{1}{d-1}|v\rangle\langle v|. \quad (28)$$

Applying this construction independently on the two local systems gives the same logical state and joint distributions. This appears to establish an adapted counterexample for every $d \geq 3$ and every prescribed local MUB pair. Because this is a new extension beyond the original qutrit certificate, it should still be checked independently before being quoted as a theorem.

The computational/Fourier family is verified numerically in the full ambient space by `scripts/verify_cqc_dimension_family.py`.

3 The two-qubit case

3.1 Exact no-go for this family

At $d = 2$,

$$\Delta_2(\epsilon) = h_2(\epsilon) - 2h_2(\epsilon/2). \quad (29)$$

Its derivative is

$$\Delta'_2(\epsilon) = \log_2 \left(\frac{1-\epsilon}{2-\epsilon} \right) < 0, \quad 0 < \epsilon < 1. \quad (30)$$

Since $\Delta_2(0) = 0$,

$$\Delta_2(\epsilon) < 0 \quad \text{for every } 0 < \epsilon \leq 1. \quad (31)$$

The structural reason is that for $d > 2$ the construction uses a proper local subspace,

$$\mathbb{C}^2 \subsetneq \mathbb{C}^d. \quad (32)$$

After compression, the ambient MUBs become less complementary logical measurements, allowing largely overlapping correlations to be counted twice. At $d = 2$, the logical support is the entire space and there is no compression mechanism.

3.2 Could a different two-qubit counterexample exist?

No such counterexample is known from the present work, and there is no proof here that none exists. Evidence favoring a surviving qubit theorem includes:

- the original search over 10^7 random $2 \otimes 2$ states and perturbations of saturating states, with no violation found [1];
- proved CQC cases including pure states, states with a maximally mixed marginal, minimally disturbing measurements, Bell-diagonal and maximally correlated states, and asymmetric Werner states [1];
- exploratory supplementary random sampling and separate multistart numerical optimization over general two-qubit density matrices, with no positive gap found. The repository includes a configurable random-search implementation at `scripts/search_cqc_qubit.py`. These calculations are informal numerical evidence only.

A reasonable working conclusion is

$$\text{CQC is false for every } d \geq 3, \quad \text{while the unrestricted } d = 2 \text{ case may survive.} \quad (33)$$

3.3 A useful qubit reduction

Fix the MUBs to Pauli Z and X , without loss of generality, and write a general two-qubit state as

$$\rho = \frac{1}{4} \left[I \otimes I + \mathbf{a} \cdot \boldsymbol{\sigma} \otimes I + I \otimes \mathbf{b} \cdot \boldsymbol{\sigma} + \sum_{i,j} T_{ij} \sigma_i \otimes \sigma_j \right]. \quad (34)$$

For $n = x, z$, the measured binary probabilities are

$$p_{st}^{(n)} = \frac{1}{4} (1 + sa_n + tb_n + stT_{nn}), \quad s, t \in \{\pm 1\}. \quad (35)$$

Thus the classical side depends only on

$$a_x, a_z, b_x, b_z, T_{xx}, T_{zz}, \quad (36)$$

whereas quantum mutual information depends on the complete positive-semidefinite completion of the Bloch-correlation data.

This suggests a proof strategy: minimize $I(A : B)$ over physical two-qubit states with those six measured moments fixed, then compare that minimum with the two observed classical mutual informations. Any two-qubit counterexample must evade every known proved case: both marginals must be nonmaximally mixed, and none of the four local observables may commute with its corresponding reduced state.

4 General upper bounds for mixed states and MUBs

Let

$$J_i = I(M_i^A : M_i^B), \quad S_m = \sum_{i=1}^m J_i, \quad (37)$$

where the m local measurement pairs are pairwise mutually unbiased in equal dimension d .

4.1 Dimension-only bound

Since $J_i \leq \log_2 d$,

$$S_m \leq m \log_2 d. \quad (38)$$

This is the optimal bound depending only on d and m , uniformly over all allowed paired MUB choices, even if the state is required to be strictly mixed. Full-rank depolarized maximally entangled states can approach a maximally entangled state arbitrarily closely. For a fixed collection of possibly mismatched local basis pairs, their cross-party geometry may permit a smaller bound.

4.2 Conditional-entropy bound

The Coles–Piani information-exclusion relation for any two MUBs gives [4, 2]

$$I(M_i^A : B) + I(M_j^A : B) \leq \log_2 d - H(A | B). \quad (39)$$

Measuring B cannot increase mutual information, so

$$J_i + J_j \leq \log_2 d - H(A | B). \quad (40)$$

Interchanging A and B gives a second bound. Summing over all $\binom{m}{2}$ pairs, with each J_i appearing $m - 1$ times, yields

$$S_m \leq \frac{m}{2} \min\{\log_2 d - H(A | B), \log_2 d - H(B | A)\}. \quad (41)$$

Equivalently,

$$S_m \leq \frac{m}{2} [\log_2 d - H(AB) + \min\{H(A), H(B)\}]. \quad (42)$$

For two MUBs,

$$I_Z + I_X \leq \min\{\log_2 d - H(A | B), \log_2 d - H(B | A)\}. \quad (43)$$

Subtracting $I(A : B)$ gives

$$I_Z + I_X - I(A : B) \leq \log_2 d - \max\{H(A), H(B)\}. \quad (44)$$

Consequently:

- if either marginal is maximally mixed, CQC holds;
- any violation requires an entropy deficit in both marginals;
- that deficit provides a rigorous state-dependent ceiling on the violation.

For qubits,

$$I_Z + I_X - I(A : B) \leq 1 - \max\{H(A), H(B)\}. \quad (45)$$

4.3 Bounds in terms of global mixedness

Using $H(A), H(B) \leq \log_2 d$ gives

$$S_m \leq m \log_2 d - \frac{m}{2} H(AB). \quad (46)$$

If only the global purity

$$\mu = \text{Tr } \rho^2 \quad (47)$$

is known, then $H(AB) \geq -\log_2 \mu$, yielding

$$S_m \leq \frac{m}{2} \log_2(d^2 \mu). \quad (48)$$

For two MUBs,

$$I_Z + I_X \leq \log_2(d^2 \mu), \quad (49)$$

and for two qubits this becomes $I_Z + I_X \leq \log_2(4\mu)$.

4.4 Outcome-distribution refinement

The quantum-memory uncertainty relation used in the original CQC analysis implies, for each MUB pair [3, 1],

$$J_i + J_j \leq I(A : B) + H(M_i^A) + H(M_j^A) - \log_2 d - H(A), \quad (50)$$

and likewise on subsystem B . Summing over all pairs gives

$$S_m \leq \frac{m}{2} I(A : B) + \min_{P=A,B} \left[\sum_{i=1}^m H(M_i^P) - \frac{m}{2} (\log_2 d + H(P)) \right]. \quad (51)$$

For two MUBs,

$$I_Z + I_X \leq I(A : B) + \min\{R_A, R_B\}, \quad (52)$$

where

$$R_A = H(Z^A) + H(X^A) - \log_2 d - H(A), \quad (53)$$

and similarly for B .

CQC asks whether the mutual information lost when the second subsystem is also measured always compensates for this residual term. The counterexample shows that it need not.

For the all- d family,

$$R_A = R_B = \log_2 \left(\frac{2(d-1)}{d} \right). \quad (54)$$

This is zero at $d = 2$ and positive for every $d > 2$, mirroring the dimensional threshold in the counterexample mechanism.

4.5 Bases that are not exactly mutually unbiased

For arbitrary orthonormal bases, let r_{ij}^A and r_{ij}^B denote the corresponding information-exclusion overlap constants. Summing the pairwise bounds gives [4, 2]

$$S_m \leq \min \left\{ \frac{\sum_{i < j} r_{ij}^A}{m-1} - \frac{m}{2} H(A | B), \frac{\sum_{i < j} r_{ij}^B}{m-1} - \frac{m}{2} H(B | A) \right\}. \quad (55)$$

For exact MUBs, every $r_{ij} = \log_2 d$, recovering the preceding result.

5 A compact follow-up research program

The calculations suggest three tightly linked projects:

1. **Formalize the $d \geq 3$ family.** Prove the computational/Fourier result in full generality, then verify or correct the arbitrary-MUB adaptation.
2. **Resolve the qubit case.** Use the six-moment reduction to formulate a constrained minimization or semidefinite-relaxation proof. A counterexample search should focus outside every known proved class.
3. **Sharpen replacement bounds.** Determine the best bound as a function of dimension, number and geometry of measurement bases, marginal entropies, global entropy or purity, and perhaps local support dimensions.

6 Current answers to the motivating questions

Higher dimensions. Yes: the computational/Fourier family gives an explicit counterexample in every $d \times d$ system with $d \geq 3$, with closed-form gap $\Delta_d(\epsilon) = h_2(\epsilon) - 2h_2(\epsilon/d)$. The same compression argument appears to adapt the construction to any prescribed local MUB pair.

Two qubits. No two-qubit counterexample is known from this work. This higher-dimensional family has an exact no-go at $d = 2$, and exploratory searches have found no alternative. The unrestricted qubit case remains open and may survive as a separate theorem.

Replacement bounds. Conditional-entropy, mixedness, purity, outcome-distribution, and basis-overlap bounds remain valid and provide a structured replacement for the failed universal inequality.

Reproducibility

The qutrit exact certificate, all-dimension verifier, and exploratory two-qubit search are in the public repository:

- `scripts/verify_cqc_exact.py`;
- `scripts/verify_cqc_dimension_family.py`;
- `scripts/search_cqc_qubit.py`.

The exact qutrit result and the algebraic formulas above do not depend on the numerical searches.

References

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